



BSc (Hons) Computer Science and Software
Engineering

Astrophotography Assistant and object identifier

Reflective Report

Student Registration Number	2425351
Student Name	O P M S P Wimalarathne
Supervisor's Name	Mrs Ovini Senevirathne
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1. Introduction

The “Mobile astrophotography assistance and celestial object recognition” project aims to bring astrophotography using smartphones and celestial object identification, more accessible among the society and amateur astronomers. Developed as part of the BSc (Hons) in Computer Science and Software Engineering program, the project tries to build a web application and a mobile application with capabilities such as recognizing and labelling the most common celestial bodies in the night sky that can be captured using a mobile smartphone with the help of Artificial Intelligence and machine learning techniques.

This project is motivated by personal interest in astronomy and astrophotography, also to bring solutions to some common difficulties and challenges faced by the public when observing the night sky. These challenges cause some difficulties such as recognizing specific celestial objects without knowledge, variability in observation conditions (light pollution, weather), and the lack of real-time guidance tools. By using neural networks, integrating image processing techniques and some astronomical databases, this system aims to build a celestial objects recognition system and enhance the night sky watching experience.

2. Self-Management

Self-management was the main secret for carrying this project forward, especially with the existing challenges and difficulties arising from the environmental and technical challenges unique to astronomical image analysis.

2.1. Self-Motivation

This project includes both practical and technical challenges. Unlike other common projects, this project completely depends on environmental conditions such as clear skies with no clouds and the moon, visibility of stars, light pollution etc. which are unique to this project. For the development of this project, it requires a lot of patience and persistence. The only key to this project becoming real was self-motivation and internal passion about space.

Several key strategies supported the maintenance of motivation:

1. **Clear Personal Goals:** Mainly this project was aligned with personal interests in astrophotography, astronomy, artificial intelligence and computer vision.
2. **Intrinsic Passion for Astronomy:** A lifelong fascination with the night sky served as a powerful driver. The idea of enabling others to learn about space using just their smartphones provided deep personal satisfaction and a larger purpose for the project.
3. **Visual and Technical Progress:** Each step forward—whether successfully integrating the camera module in React Native or identifying even one star through trial datasets—reinforced confidence. Celebrating these small wins helped overcome the long gaps in observational data.
4. **Backup Plans for Bad Conditions:** Acknowledging that weather could interrupt image collection, the plan was intentionally flexible. UI design, synthetic dataset research, or exploring NASA open datasets were chosen as productive alternatives during downtime.

Furthermore, finding solutions in space since childhood using AI strongly influenced this project. By making a simple but strong tool that allows the public to identify celestial objects in the night sky using your own smartphone is a great achievement for everyone who loves astronomy.

2.2.Time Management

At the beginning of the project, a clear project plan was created by breaking down the full scope into smaller components for better understanding and execution. Development in both front end and back end, Creation of datasets, training the machine learning model and testing are some of the components mentioned above.

Time blocking was used to stay organized and focused. Time slots were used for coding, documentation, UI/UX, and image processing. Weekends and nights were used only for sky observation and photography. When the field work was not possible, the time was spent testing the models on old images and developing algorithms.

Project milestones and deadlines, including sprint plans for features and thesis work, were monitored using a Gantt chart. It was updated frequently to account for data collection delays. Currently in Sprint 4, the project, which used an Agile Scrum methodology, is concentrating on giving the mobile app camera access and training the model with labeled data.

Overall, even though some dependencies caused inevitable delays, the project was able to successfully meet its academic milestones and continue to evolve thanks to strategic time management that was founded on adaptability, flexibility, and clear prioritization.

2.3.Readiness to Improve

A strong drive to grow and learn was essential to this project's success. Many challenges were overcome by remaining open-minded and growth-focused, as it was difficult to capture and identify stars in various weather and lighting conditions. The tactics employed during the project demonstrated a persistent attempt to adjust and get better at each phase.

1. Seeking Feedback: Continuous help, guidance and the feedback actively given from the supervisor, Mrs. Ovini senevirathne, as well as peers and the public. Those helped to enhance the quality of the project.
2. Reflective Learning: Throughout the project, a digital journal was maintained to document daily tasks, technical difficulties, model experiments, and individual reflections. This made it easier to keep track of what worked, what didn't, and what needed to be altered. The journal, for instance, pointed out issues with using star data from various times and perspectives, which improved data collection organization.
3. Adaptability: The project wanted to have a high adaptability due to some conditions like weather, device limitations, moonlight and technical. For example, when collecting images for dataset during rapidly changing weather or when the results of the model was unsatisfied, alternative models and techniques were researched and used.
4. Skill Development: Throughout the project, learning persisted. I studied subjects like computer vision, mobile development, and astronomy image processing using YouTube, online tutorials, and

courses. I investigated things like TensorFlow Lite for mobile, star position mapping, and enhancing night sky photos. These abilities supported my development as a developer and enhanced the project.

2.4.Summary

The development of mobile astrophotography assistance and object identifier was guided by strong self-management. By setting simple and small goals, they helped managing tasks. For steady progress, time management tools such as Gantt chart and agile methods were helped. Regular feedback and reviews were useful in development and in the improvement of the system, it also encouraged learning.

3. Reflection

This section offers a critical analysis of the Celestial Object Recognition and Labeling System's current development, assessing how well it aligns with the initial goals and benchmarks. Additionally, it looks at the main obstacles faced, the methods used to overcome them, the lessons discovered, and the areas that need improvement in the future.

3.1.Current Overview of the Project Process

A. Project Planning, Requirement Gathering & Analysis:

The project began with a well-defined plan to develop a mobile application that uses AI to identify and categorize stars. Agile Scrum was utilized to manage development in phases, and a schedule and resources were established. Using the mobile camera, connecting to star maps, executing a machine learning model on the device, and displaying real-time labels on the screen were among the essential requirements.

B. System Design & Implementation:

A modular system that recognizes stars accurately was created to function well on mobile devices. Real-time labeling, star matching, GPS and sensor use, and picture capture were important components. Figma was used to create UI designs for a seamless user experience. React Native, TensorFlow Lite, and star data from programs like Stellarium or Skyfield were used in the system's construction.

C. Testing & Refining the System:

Every component, particularly the camera, real-time sky overlays, and model predictions, underwent unit and integration testing. Every sprint included testing to gather feedback and address problems. To see how the system functions in practice and adjustment, we also tested challenging scenarios, such as taking photos in cloudy conditions or with incorrect orientation data. The system became faster, more accurate, and easier to use because of this repeated process.

D. Documentation:

Comprehensive documentation that contains technical details as well as user manuals. The setup, usage, system architecture, and deployment of the application are all covered. For the benefit of both technical teams and users, the documentation is updated frequently to reflect new developments and expansion plans.

Challenges & Issues, and Solutions for Each of the Above Stages

Stage	Challenges & Issues	Solutions
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A	• Limitations in hardware caused changes in the scope. • Image capturing time, location and the weather made requirements unpredictable	• Conducted several tests under different conditions to define practical goals • Focused on basic and common features of all mobile devices
B	• Real time processing slowed the performance on low and medium end devices	• Used optimized and lightweight models for better performance
C	• Accuracy of star recognition dropped due to unexpected weather conditions and cloudy skies	• Improved model by adding varied sky images to training data.
D	• Technical terms were difficult for general users	• Created different documentation versions for both simple and extended

Table 3.1: Challenges & Issues and Solutions

3.2.Current Status of Development

The project has made good progress, and it is on the way to completion. The following are the major achievements so far achieved

1. Sky Image Capture and Preprocessing:
Users can capture Skype images in real time by using their mobile camera. Preprocessing steps like light pollution correction, adjustment in contrast, and image cropping have been used
2. Star Pattern Recognition and Matching:
A lightweight machine learning model has been in the system to identify objects and star patterns from the captured images. Due to the limitations in mobile cameras, it is extremely efficient and effective
3. Live updates and news
the system is also informed about the upcoming events and the news related to space and astronomy.
4. Optimized settings:
The data which used in the model were captured by different kinds of mobile devices from low end to high end. Therefore, the system provides optimized average settings for several devices.

3.3.Functions to be Completed

Despite the substantial progress, a few critical tasks remain:

- **Increase the accuracy of the model:**

The current model Performs well with few bright and some major planets or stars. For better performance the accuracy can be further increased by populating the data set with more detailed and diverse images which are taken under different weather conditions, angles and lightning conditions.

- **Real time labeling:**

Real time celestial objects labeling through the camera remains as a key feature. This need to be done with the real time and a minimal delay in using device sensors such as GPS gyroscope and the compass.

- **Comprehensive Testing:**

- **User Acceptance Testing (UAT):** By enrolling small group of users to test basic features and good feedback
- **Automated Testing:** Implementing test scripts for UI responsiveness and backend functions to ensure repeatable, efficient validations.
- **Cross-Platform Compatibility:** By using several devices with different screen sizes, operating systems, and different environmental conditions

3.4.Challenges & Issues Faced in Development, and Solutions

As this project was being developed, several logistical and technical issues came up. However, these difficulties were effectively resolved through strategic planning, ongoing education, and efficient problem-solving, allowing the project to move forward.

1. **Identifying Celestial Objects Accuracy:** Environmental factors like light pollution, atmospheric haze, and different image resolutions made it difficult to achieve high accuracy in star recognition.
 - **Solution:** The model was trained on data that represented a variety of sky conditions, and preprocessing filters were applied to improve image contrast. Furthermore, a fallback method was included to cross-validate object predictions using GPS and compass data.
2. **Frontend and Backend Communication Delays:** Latency problems were caused by sporadic delays in synchronizing real-time camera input with backend star data.
 - **Solution:** To avoid UI blocking and offer a more seamless user experience, asynchronous data handling techniques and local caching of recent predictions were used.

By strengthening the system's resilience and dependability, these fixes have set it up for final testing and implementation. Every challenge offered worthwhile educational opportunities that improved technical proficiency and project management skills.

3.5.Personal Development and Lessons Learned

Key Personal Development Skills include:

- **Advanced Technical Competency**

Working with object detection, real time augmented reality, machine learning, sensor data and GPS has strengthened my knowledge and understanding of how these different technologies work together. I learned to use optimization, lightweight models, And image recognition. Those experiences improved my knowledge and technical problems solving skills.

- **Improved Project Management**

Using agile scrum Helped me in managing the project More effectively when breaking tasks. This improved flexibility, planning and accountability. Regular reviews and tools like task tracking helped me to stay on schedule.

- **Incorporating Feedback and Effective Communication**

Frequent feedback sessions, which included discussions with supervisors and self-evaluations, were crucial for improving the project's functionality and design. Transforming abstract objectives into useful technical features was made easier by open and transparent communication.

Key lessons learned include:

- **Clear Requirement Definition Is Critical**

Uncertain feature expectations at the beginning of the project led to rework. This demonstrated how crucial it is to set clear, testable requirements early in the project lifecycle. This clarity was successfully attained using visual prototypes and user stories.

- **Iterative Development Drives Quality**

Early problem identification was aided by short cycles of testing and continuous development. Iterative refinement allowed for consistent progress and integrated insightful feedback for continuous enhancements rather than trying to perfect a feature in a single attempt.

- **Testing Cannot Be Overlooked**

Static testing methods were unable to identify problems that were discovered through real-world testing in a variety of devices and conditions. Delivering a stable and dependable product required extensive testing, which included edge cases, environmental simulations, and user behavior analysis.

3.6.Possible Future Modifications

Reflecting on the project, there are several potential modifications and improvements that could be considered for future iterations:

- **Extended Research Phase**

By collecting more accurate data and spending more time on research and prototyping Could increase the quality and the accuracy of the development process it will help in labeling, displaying and astronomy related calculations

- **Enhanced Peer and Industry Collaboration**

By working with professionals, experts, teachers and peers could enhance the features like

cancellation mapping, visibility predictions and the accuracy of stars and planets. Their inputs and help make the application much more accurate, practical and educational

- Early and Continuous User Testing
By collecting ideas and feedback by involving users like amateur astronomers, students and the public in the earlier stages Could help to identify real world needs
- Advanced AR and Recognition Features
Future versions can have advanced features such as:
 - Dynamic Sky Simulation: Predicting sky views for upcoming dates
 - Offline Star Maps: Allowing user to use the application in offline
 - Augmented Reality Enhancements: Improving augmented reality with real time labeling
- Improved User Interface and Experience
Potential upgrades include:
 - Customizable views (e.g., filter by object type: planets, stars, satellites).
 - Enhancements of night mode for better stargazing in low light.
 - Interactive tutorials for beginners.
- Model Optimization and Expansion
Future updates could integrate:
 - Lighter, faster ML models for real-time star detection on low-end devices.
 - Real-time accuracy calibration using device sensors and atmospheric conditions.
- Increased Accessibility and Language Support
Adding features like audio output, larger text, voice guidance and multilingual support to increase the accessibility.

3.7.Summary

The mobile astrophotography assistant an object identifier system has made good progress with working key features such as object detection, back-end integration, user friendly frontend for multiple devices. Asked the project reaching completion, The focus is on the accuracy of model and improving testing. The lesson learned and the gained experience Will be a solid foundation for the future upgrades and the deployment of the application commercially.

4. In-Depth Contents of the Thesis

This part provides a synopsis and outline of the thesis report. It defines the main components and chapters, giving a roadmap for navigating the in-depth analysis of the project—from its introduction and methodology to its findings, evaluation, and conclusions.

Title Page:

The university name and logo, the title of the degree, the report title, the project title, my name, and my student number will be included on the first page of the report.

Abstract:

This section will provide a brief overview of the project, synthesizing the major aims, approach, extent, and conclusions of the celestial object detection and sky mapping application creation.

Acknowledgment:

This section will acknowledge the organizations and people that took part in the project.

Table of Contents:

All sub-sections, sections, and chapters of the report.

Chapter 01: Introduction

This section will introduce the project with regard to its purpose, relevance, and technologies involved in celestial objects' identification and augmented reality sky mapping. It will also give an overview of the layout of the thesis.

- 1.1 Project Background
- 1.2 Aim and Objectives
- 1.3 Artefact Description (Sky Mapping App)
- 1.4 Project Structure

Chapter 02: Literature Review

Literature Review within this chapter will collate research and technologies available to date regarding astronomical object identification, image processing, real-time overlays, and mobile AR applications. It will discuss relevant machine learning algorithms, sensor integrations, and solutions available.

- 2.1 Introduction
- 2.2 Existing Systems and Apps
- 2.3 Technologies Utilized (Computer Vision, Machine Learning, AR)
- 2.4 Benefits of the Developed App
- 2.5 Challenges in User Experience
- 2.6 Comparative Analysis
- 2.7 Research Gaps
- 2.8 Market and User Needs Analysis

Chapter 03: Methodology

The methodology of the project shall be presented in this chapter and along with that, the Agile development process, software tools, frameworks, and data handling habits will be discussed. The chapter will also elaborate on the techniques of system designing, testing strategies, and assessment measures.

- 3.1 Selected Methodology and Reason
- 3.2 Requirement Gathering
- 3.3 System Designing
- 3.4 Implementation
- 3.5 Testing (Unit, Integration, Real-world)
- 3.6 Evaluation

Chapter 04: System Design and Implementation

This chapter will define the system architecture, hardware and software elements, data flow, and user interface. Implementation specifics and code snippets relating to the sky mapping feature will be provided as well.

- 4.1 Technology Choice
- 4.2 Backend Coding (Data Processing, APIs)
- 4.3 Frontend Coding (Real-time Camera Integration, UI)
- 4.4 Algorithm Coding (Star Identification, Sensor Integration)
- 4.5 User Interface and Experience Design

Chapter 05: Results, Discussions & Evaluation

This chapter will present test outcomes, analyze expected versus actual outcomes, and evaluate the performance of the system regarding accuracy, responsiveness, and usability. It will also denote deficiencies and areas where it needs to improve.

- 5.1 Data Analysis
- 5.2 System Performance Evaluation
- 5.3 Existing Solution Comparison
- 5.4 Findings and Insights Discussion

Chapter 06: Conclusion and Future Work

The conclusion will summarize the success of the project and its contribution to mobile astronomy applications. It will also put forward possibilities for future development and research.

- 6.1 Summary of Key Outcomes
- 6.2 Achievement of Objectives
- 6.3 Project Limitations
- 6.4 Future Work and Recommendations

Appendices

Other documents like system architecture figures, user guides, datasets, test results, code listings, and algorithm documentation will be included here.

References

A complete list of academic articles, books, online sources, and softwares quoted in the thesis report.

5. Appendix

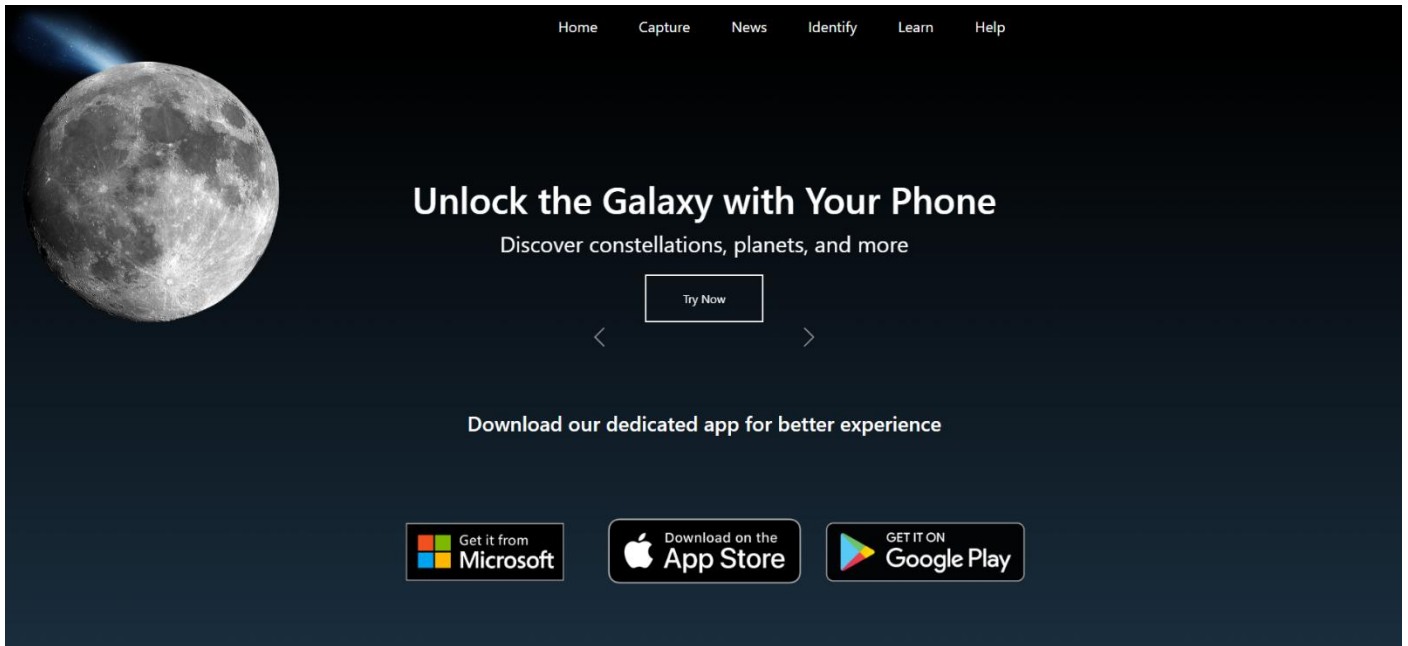


Figure 1: Web application - Home

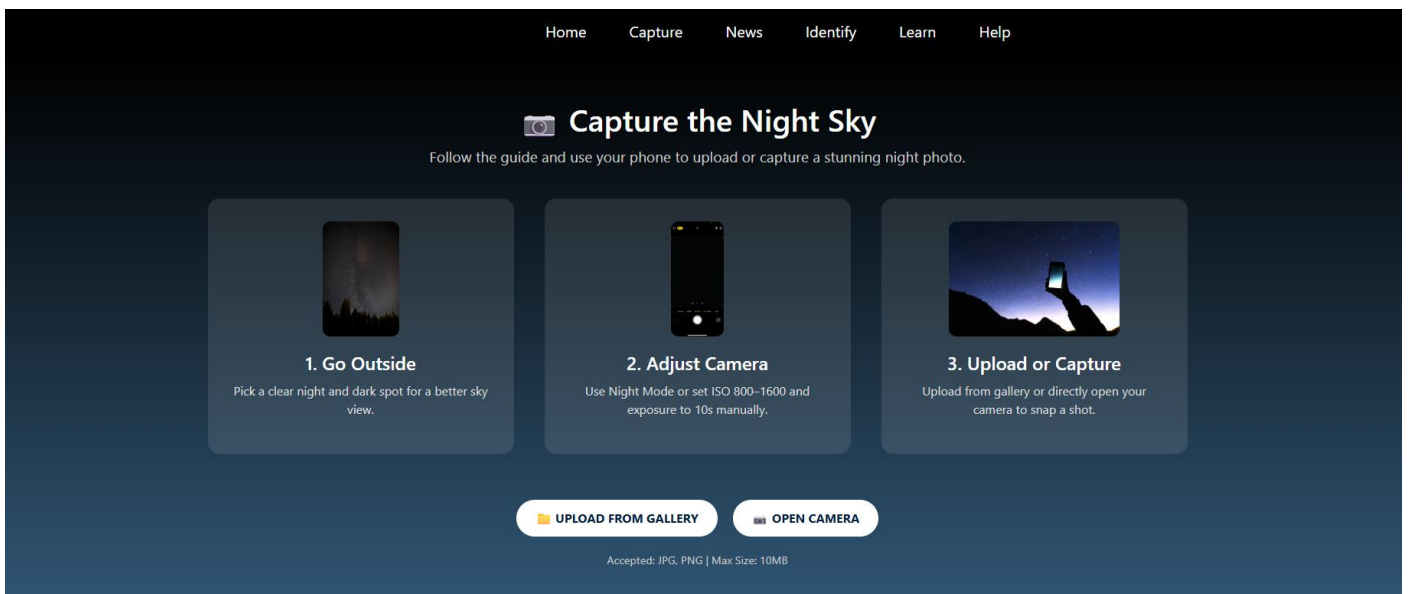
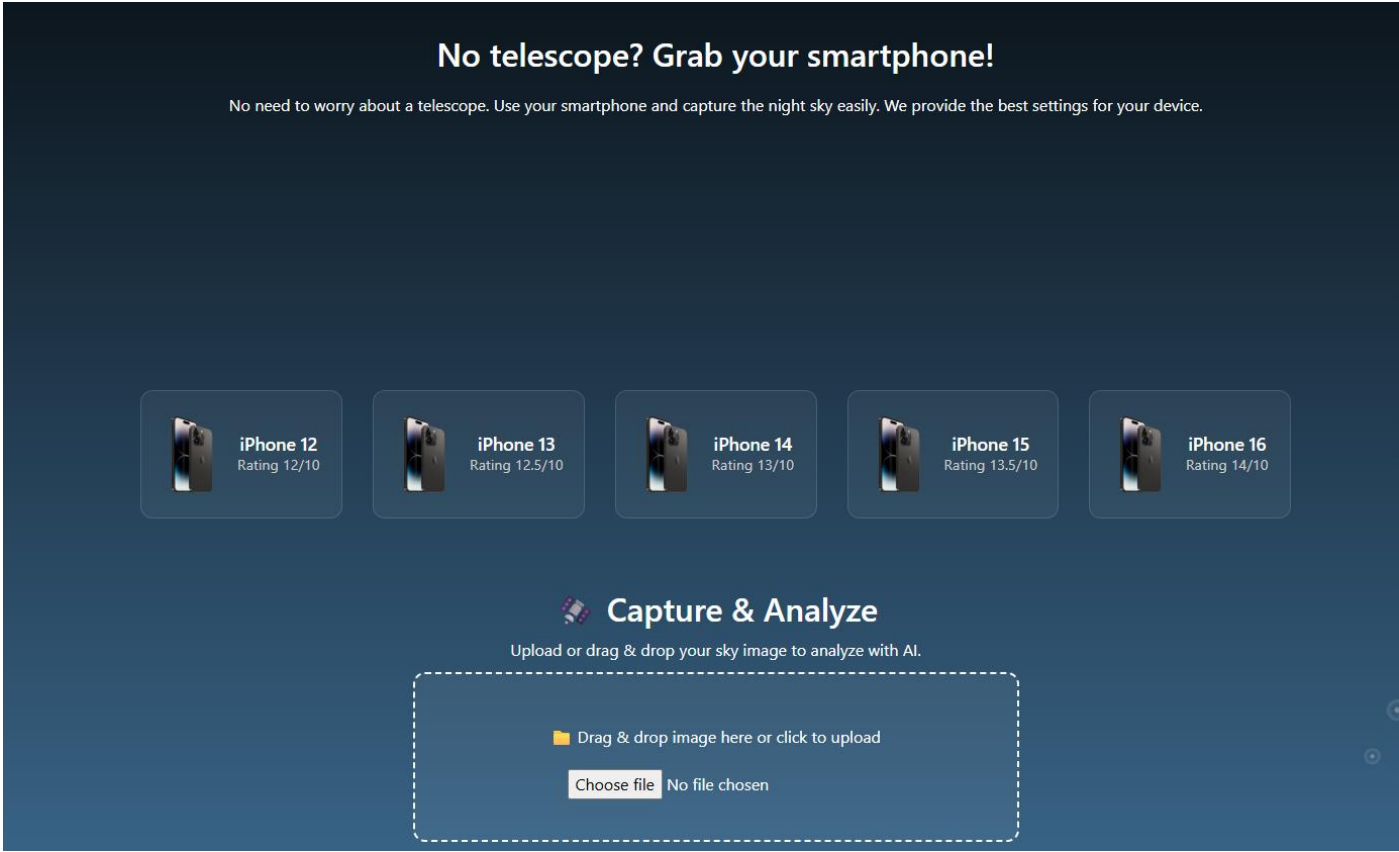


Figure 2: Web application - Capture



Figurer 3: Web application - Analyze



Figure 4: Model

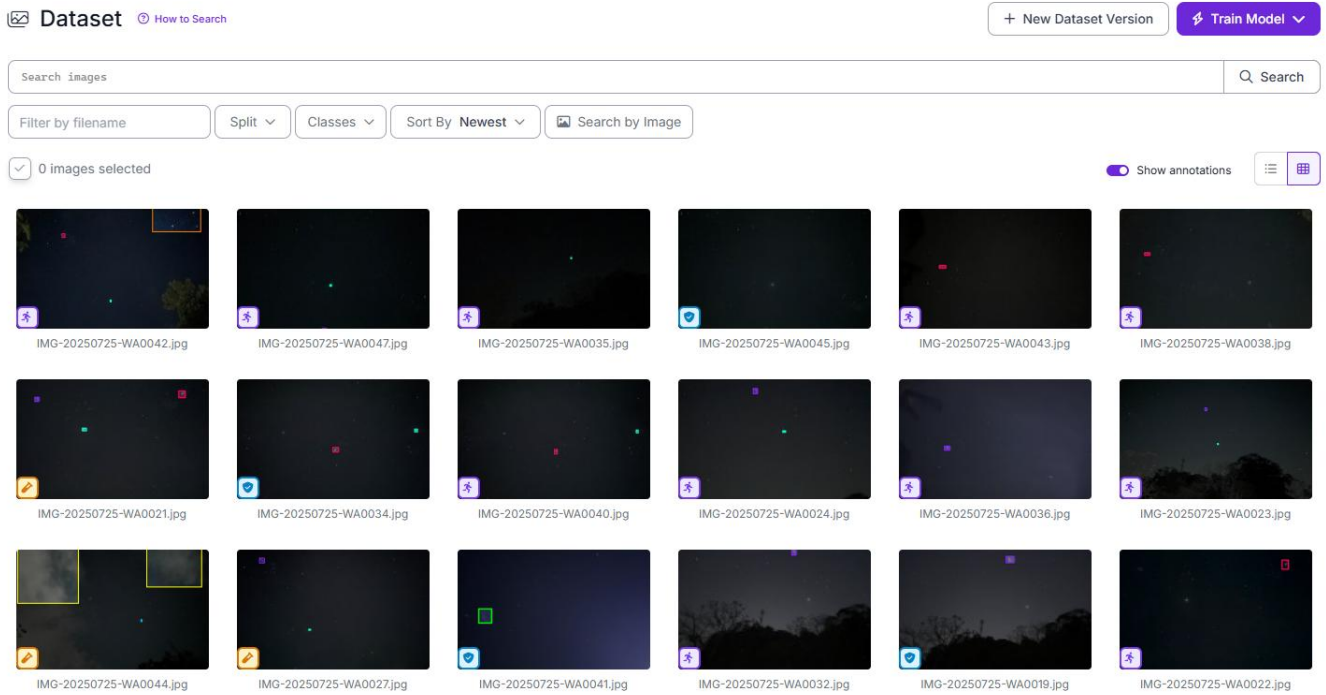


Figure 5: Model dataset